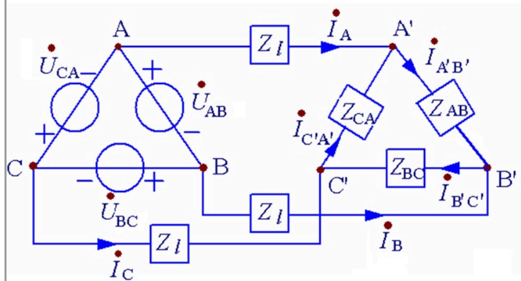
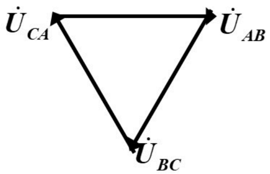
二、Δ-Δ形联接

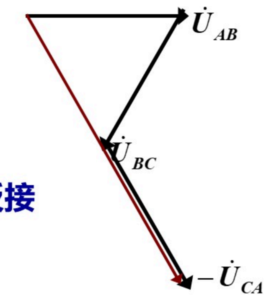


当三相电源对称时：

$$\dot{U}_{AB}+\dot{U}_{BC}+\dot{U}_{CA}=0$$



只有对称三相电源才可以接成三角形，且每一相不能反接。



设*C*相反接

单选题 1分

1. Δ－Δ 对称三相联接时线电流和负载端相电流的关系是

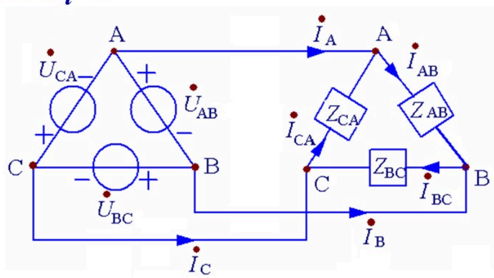
- A 线电流大小是相电流大小的√3倍，相位角超前相电流30度
- B 相等
- C 线电流大小是相电流大小的√3倍，相位角滞后对应的相电流30度

单选题 1分

2. Δ－Δ 联接时电源端线电压和相电压的关系是

- A 线电压大小是相电压大小的√3倍，相位角超前对应相电压30度
- B 相等
- C 线电压大小是相电压大小的√2倍，相位角超前对应相电压30度

1. $Z_l=0$



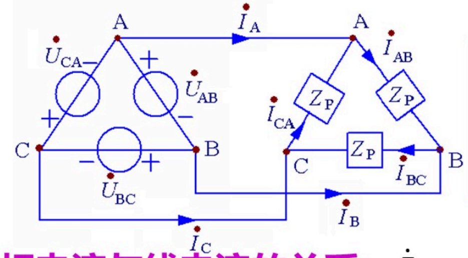
$$\boxed{\dot{U}_p=\dot{U}_l}$$

相电流
$$\begin{cases}\dot{I}_{AB}=\dfrac{\dot{U}_{AB}}{Z_{AB}}\\ \dot{I}_{BC}=\dfrac{\dot{U}_{BC}}{Z_{BC}}\\ \dot{I}_{CA}=\dfrac{\dot{U}_{CA}}{Z_{CA}}\end{cases}$$

线电流
$$\begin{cases}\dot{I}_A=\dot{I}_{AB}-\dot{I}_{CA}\\ \dot{I}_B=\dot{I}_{BC}-\dot{I}_{AB}\\ \dot{I}_C=\dot{I}_{CA}-\dot{I}_{BC}\end{cases}$$

先算相电流，再算线电流

负载对称（$Z_{AB}=Z_{BC}=Z_{CA}=Z_p$）



$$\dot{I}_{AB}=\frac{\dot{U}_{AB}}{Z_p}$$

$$\dot{I}_{BC}=\frac{\dot{U}_{BC}}{Z_p}$$

$$\dot{I}_{CA}=\frac{\dot{U}_{CA}}{Z_p}$$

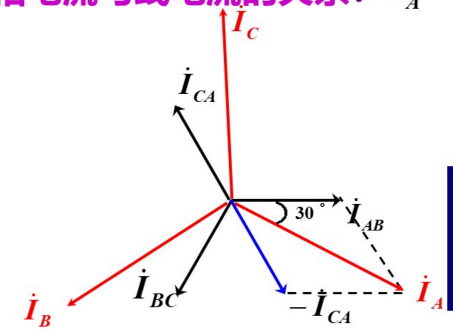
$\dot{I}_{AB}$、$\dot{I}_{BC}$、$\dot{I}_{CA}$ 对称

$\dot{I}_{A}$、$\dot{I}_{B}$、$\dot{I}_{C}$ 对称

相电流与线电流的关系：$\dot{I}_A=\dot{I}_{AB}-\dot{I}_{CA}=\sqrt{3}\dot{I}_{AB}\angle-30^\circ$

$$\dot{I}_B=\sqrt{3}\dot{I}_{BC}\angle-30^\circ$$

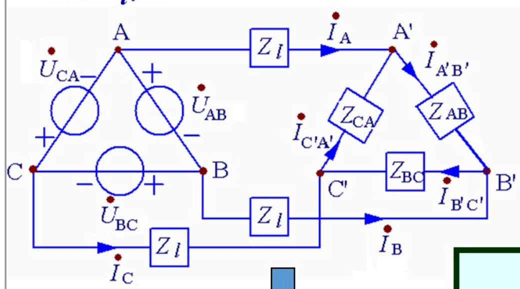
$$\dot{I}_C=\sqrt{3}\dot{I}_{CA}\angle-30^\circ$$



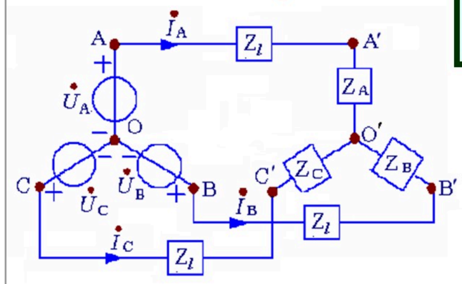
$$I_l=\sqrt{3}I_p$$

线电流滞后对应相电流30°

2. $Z_l\neq0$



三相电源和三相负载均进行Δ→Y变换



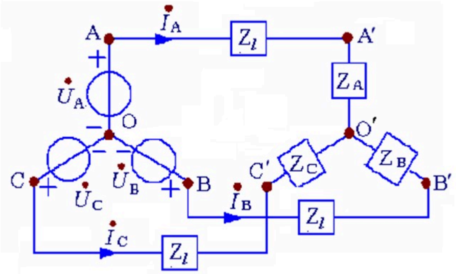
$$\dot{U}_A=\frac{\dot{U}_{AB}\angle-30^\circ}{\sqrt{3}}$$

$\dot{U}_B=\dot{U}_A\angle-120^\circ$，$\dot{U}_C=\dot{U}_A\angle120^\circ$

$$Z_A=\frac{Z_{AB}Z_{CA}}{Z_{AB}+Z_{BC}+Z_{CA}}$$

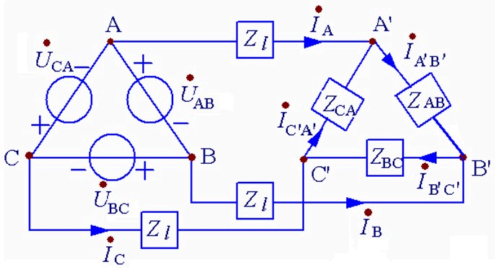
$$Z_B=\frac{Z_{BC}Z_{AB}}{\sum Z}\qquad Z_C=\frac{Z_{CA}Z_{BC}}{\sum Z}$$

$$\dot{U}_{o'o}=\frac{\dfrac{\dot{U}_A}{Z_A+Z_l}+\dfrac{\dot{U}_B}{Z_B+Z_l}+\dfrac{\dot{U}_C}{Z_C+Z_l}}{\dfrac{1}{Z_A+Z_l}+\dfrac{1}{Z_B+Z_l}+\dfrac{1}{Z_C+Z_l}}$$



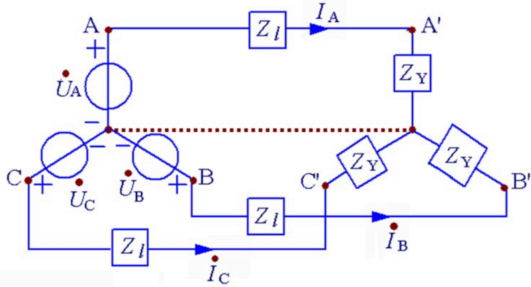
线电流

$$\begin{cases}\dot{I}_A=\dfrac{\dot{U}_A-\dot{U}_{o'o}}{Z_A+Z_l}\\[2ex]\dot{I}_B=\dfrac{\dot{U}_B-\dot{U}_{o'o}}{Z_B+Z_l}\\[2ex]\dot{I}_C=\dfrac{\dot{U}_C-\dot{U}_{o'o}}{Z_C+Z_l}\end{cases}$$



$$\left.\begin{aligned}\dot{I}_A&=\dot{I}_{AB}-\dot{I}_{CA}\\\dot{I}_B&=\dot{I}_{BC}-\dot{I}_{AB}\\\dot{I}_C&=\dot{I}_{CA}-\dot{I}_{BC}\end{aligned}\right\}\Rightarrow$$ 相电流

负载对称（$Z_{AB}=Z_{BC}=Z_{CA}=Z_\Delta$）



$$Z_Y=\frac{1}{3}Z_\Delta$$

$$\dot{I}_A=\frac{\dot{U}_A}{Z_l+Z_Y}$$

$$\dot{I}_B=\dot{I}_A\angle-120^\circ$$

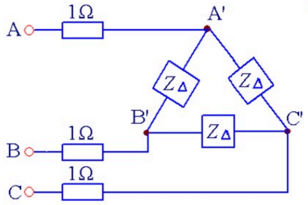
$$\dot{I}_C=\dot{I}_A\angle120^\circ$$

$$\dot{I}_{AB}=\frac{\dot{I}_A\angle30^\circ}{\sqrt{3}}$$

$\dot{I}_{BC}=\dot{I}_{AB}\angle-120^\circ$，$\dot{I}_{CA}=\dot{I}_{AB}\angle120^\circ$

注意：线电流、相电流

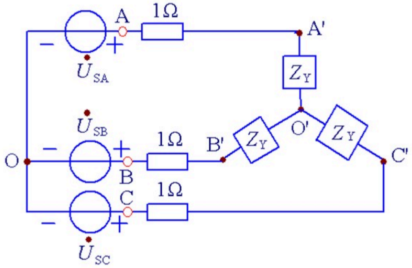
例1. 下图所示三相电路中，对称三相负载作Δ联接，每相阻抗Z_Δ=（24+j36）Ω，线路电阻为1 Ω。ABC三端接至线电压为380 V的对称三相电源。求负载各相电流相量和各线电流相量。



解：令 $\dot U_{SA} = 220\angle 0^\circ$ V

$$Z_Y = \frac{1}{3} Z_\Delta = \frac{1}{3}(24+j36)\,\Omega = (8+j12)\,\Omega$$

三相化单相计算

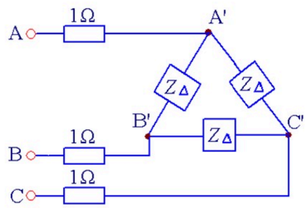


$$\dot I_A = \frac{\dot U_{sA}}{1+(8+j12)} = \frac{220\angle 0^\circ}{15\angle 53.13^\circ}\ \text{A} = 14.66\angle -53.13^\circ\ \text{A}$$

$$\dot I_B = 14.66\angle -173.13^\circ\ \text{A}$$

$$\dot I_C = 14.66\angle 66.87^\circ\ \text{A}$$

$$\dot I_A = 14.66\angle -53.17^\circ\ \text{A}$$



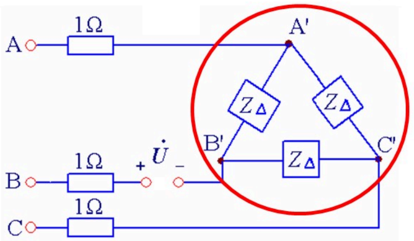
各相相电流

$$\dot I_{A'B'} = \frac{\dot I_A\angle 30^\circ}{\sqrt{3}} = 8.46\angle -23.17^\circ\ \text{A}$$

$$\dot I_{B'C'} = 8.46\angle -143.17^\circ\ \text{A}$$

$$\dot I_{C'A'} = 8.46\angle 96.83^\circ\ \text{A}$$

例2. 例1所示电路中，如B线因故断开，求负载各相电流相量和各线电流相量，以及断开处的电压。



解：令 $\dot U_A = 220\angle 0^\circ$ V

$\dot U_{AB} = 380\angle 30^\circ$ V

$$Z = \frac{2Z_\Delta Z_\Delta}{3Z_\Delta} = \frac{2}{3}(24+j36) = 16+j24$$

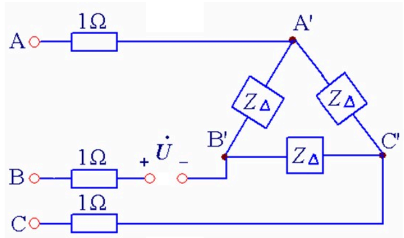
$$\dot I_A = \frac{\dot U_{AC}}{1+1+Z} = \frac{380\angle -30^\circ}{18+j24} = 12.67\angle -83.13^\circ\text{A}$$

$\dot I_C = -\dot I_A = 12.67\angle 96.87^\circ$A　　$\dot I_B = 0$

由分流公式可得

$$\dot I_{A'B'} = \dot I_{B'C'} = \frac{1}{3}\dot I_A = 4.22\angle -83.13^\circ\text{A}$$

$$\dot I_{C'A'} = \frac{2}{3}\dot I_C = 8.45\angle 96.87^\circ\text{A}$$



由KVL可得

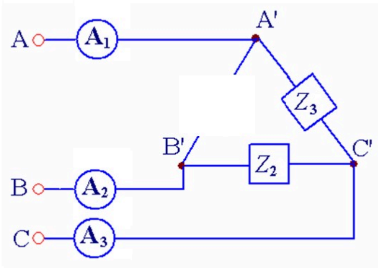
$$\dot U + Z_\Delta \dot I_{B'C'} - 1\dot I_C = \dot U_{BC} \Rightarrow$$

$$\dot U = \dot U_{BC} + \dot I_C - Z_\Delta \dot I_{B'C'}$$

$$= 380\angle -90^\circ + 12.67\angle 96.13^\circ - (24+j36)\cdot 4.22\angle -83.13^\circ$$

$$= 329.18\angle -119.98^\circ\ \text{V}$$

例3. 下图所示对称三相电路中，$Z_1=Z_2=Z_3$，三个电流表的读数均为10A。断开Z_1，求各电流表的读数。



解：

Z_1断开前各电流表的读数为线电流

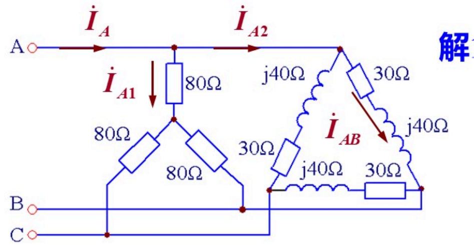
Z_1断开后

A_1、A_2的读数为相电流

$$\frac{10}{\sqrt{3}}=5.77\text{A}$$

$\dot{I}_C=\dot{I}_{C'A'}-\dot{I}_{B'C'}$保持不变，故$A_3$=10

例4. 下图表示一线电压为380V，同时接有星形和三角形负载的对称三相电路，试求输电线上的电流。



解1：令 $\dot{U}_A=220\angle 0^\circ\text{V}$

则 $\dot{U}_{AB}=380\angle 30^\circ\text{V}$

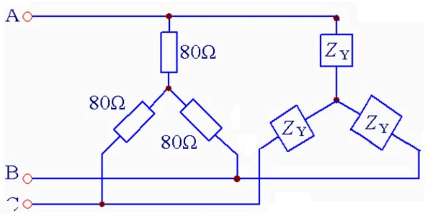
$$\dot{I}_{A1}=\frac{\dot{U}_A}{80}=\frac{220\angle 0^\circ}{80}=2.75\angle 0^\circ\text{A}$$

$$\dot{I}_{AB}=\frac{\dot{U}_{AB}}{30+\text{j}40}=\frac{380\angle 30^\circ}{30+\text{j}40}=7.6\angle -23.1^\circ\text{A}$$

$$\dot{I}_{A2}=\sqrt{3}\dot{I}_{AB}\angle -30^\circ=13.16\angle -53.1^\circ\text{A}$$

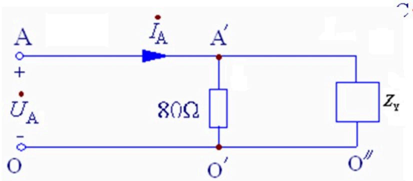
$$\dot{I}_A=\dot{I}_{A1}+\dot{I}_{A2}=2.75\angle 0^\circ+13.16\angle -53.1^\circ=15\angle -44.7^\circ A$$

解2：Δ→Y



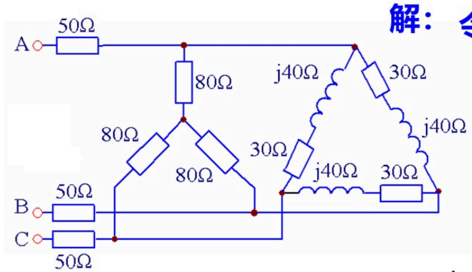
$$Z_Y=\frac{1}{3}Z_\Delta=\frac{1}{3}(30+\text{j}40)\Omega$$

$$=\frac{50}{3}\angle 53.1°\Omega$$



$$\dot{I}_A=\frac{\dot{U}_A}{Z_A}=\frac{220\angle 0°}{\dfrac{80\times\dfrac{50}{3}\angle 53.1°}{80+\dfrac{1}{3}(30+j40)}}=15\angle -44.7°\text{ A}$$

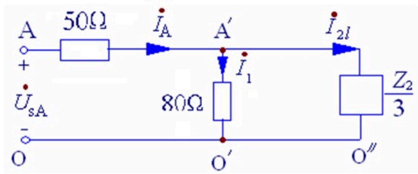
例5 下图表示一线电压为380V，同时接有星形和三角形负载的对称三相电路，分别求输电线上的电流、两组负载的相电流。



解：令 $\dot{U}_{sA}=220\angle 0^\circ\text{V}$

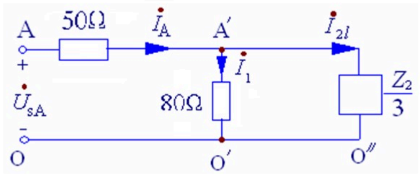
$$Z_Y=\frac{1}{3}Z_\Delta=\frac{1}{3}(30+\text{j}40)\Omega$$

$$=\frac{50}{3}\angle 53.1°\Omega$$



$$\dot{I}_A=\frac{\dot{U}_{sA}}{Z_A}=\frac{220\angle 0°}{50+\dfrac{80\times\dfrac{50}{3}\angle 53.1°}{80+\dfrac{1}{3}(30+j40)}}$$

$$=\frac{220\angle 0°}{61.29\angle 9.68°}=3.59\angle -9.68°\text{A}$$



$\dot{I}_A = 3.59\angle -9.68°\text{A}$

由分流公式可得

$$\dot{I}_1 = \frac{\frac{50}{3}\angle 53.1^\circ}{80 + \frac{1}{3}(30 + j40)} \times 3.59\angle -9.68^\circ = 0.66\angle 35^\circ\ \text{A}$$

$$\dot{I}_{2l} = \frac{80}{80 + \frac{1}{3}(30 + j40)} \times 3.59\angle -9.68^\circ = 3.16\angle -18.11^\circ\ \text{A}$$

$$\dot{I}_{2p} = \frac{\dot{I}_{2l}\angle 30^\circ}{\sqrt{3}} = 1.82\angle 11.89^\circ\ \text{A}$$

投票 最多可选1项

3. Δ－Δ联接端线上阻抗Z_l=0时，**通常计算时可**

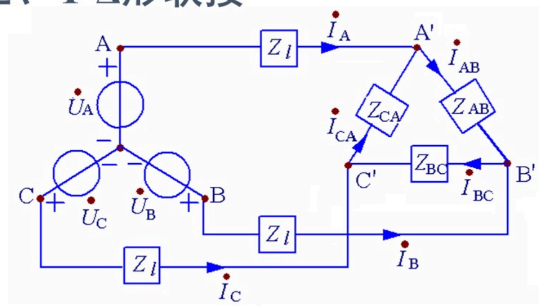
- A 先计算出线电流，再计算负载端相电流
- B 先计算出负载端相电流，再计算线电流
- C 不知道

单选题 1分

8. Δ－Δ联接端线上阻抗Z_l≠0时，**通常将其化作**Y-Y联接，此时

- A 先计算出线电流，再计算负载端相电流
- B 先计算出负载端相电流，再计算线电流
- C 不知道

三、Y-Δ形联接



四、Δ-Y形联接